Dimensions and Areas – General Data

1. Aircraft Dimensions and Ground Clearances

With the aircraft supported on its landing gear, refer to Figure 1, the principal dimensions and ground clearances are as follows:

Note Dimensions to Ground Line are approximate only and vary depending on aircraft configuration and loading conditions.

(1) Overall Dimensions

Span	65 ft – 0 in.
Length	51 ft – 9 in.
Height (tires normal)	19 ft – 9 in.
Propeller ground clearance (static)	5 ft – 0 in.
Design gross weight	12,500 pounds

(2) Wings

Type	High
Chord	6 ft – 6 in.
Incidence	2 – 1/2°
Dihedral	3°
Aspect Ratio	10.0
Airfoil Section	High Lift Section

(3) Horizontal stabilizer and elevators

Span	20 ft – 8 in.
Chord	4 ft – 9 in.
Incidence	0°
Dihedral	0°
Aspect ratio	4.35
Airfoil Section	63_A213 Inverted and Modified

(4) Vertical stabilizer

Height	13 ft – 6 in.
Root Chord	10 ft – 0 in. (Basic)
Tip Chord	5 ft – 0.6 in.
Aspect ratio	1.84

(5) Fuselage

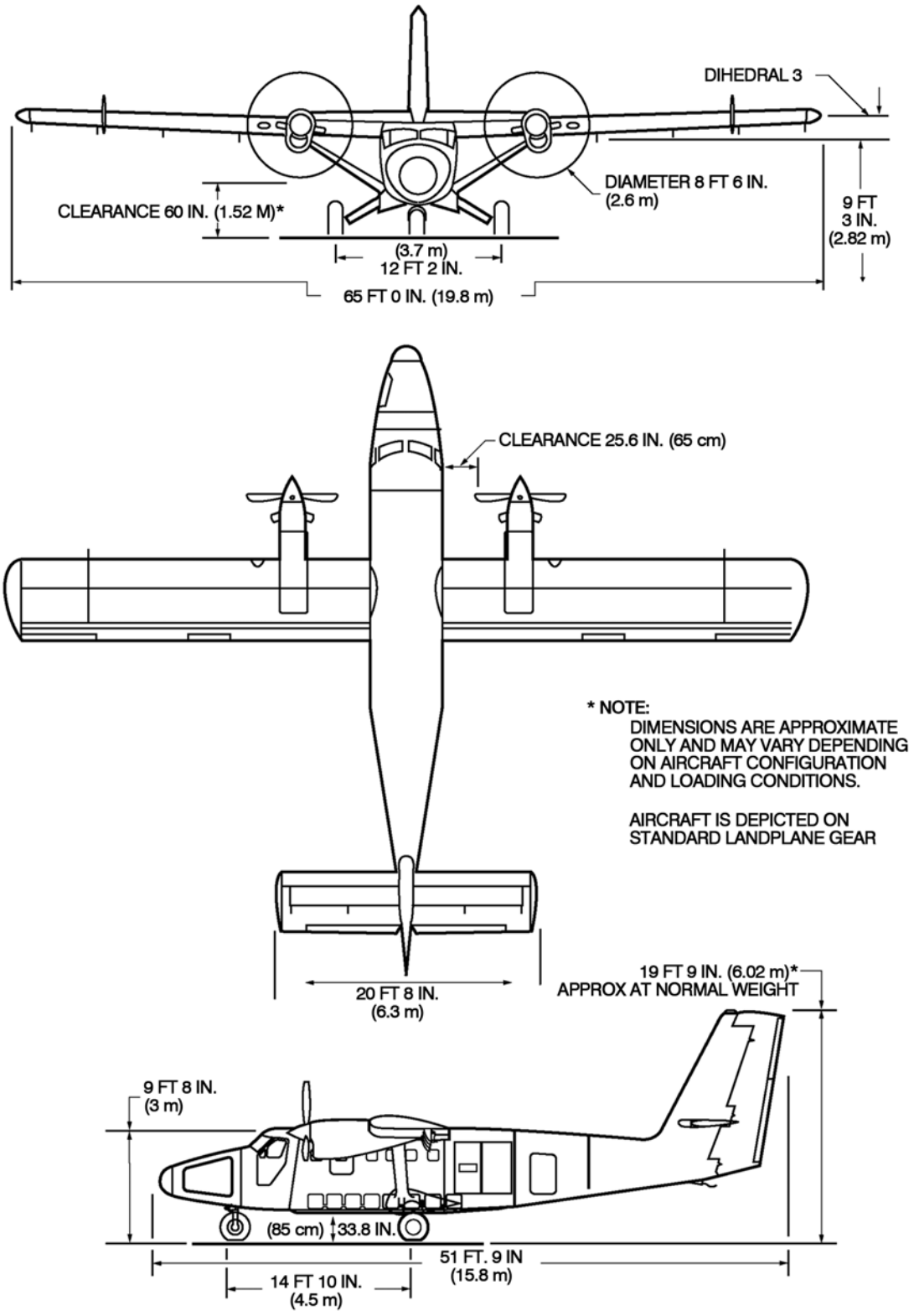
Width (maximum)	5 ft – 9.2 in.
Height (top of center fuselage)	9 ft – 8 in.

(6) Areas

Wing	420 sq ft
Horizontal stabilizer and elevators	98 sq ft

(7) Landing gear

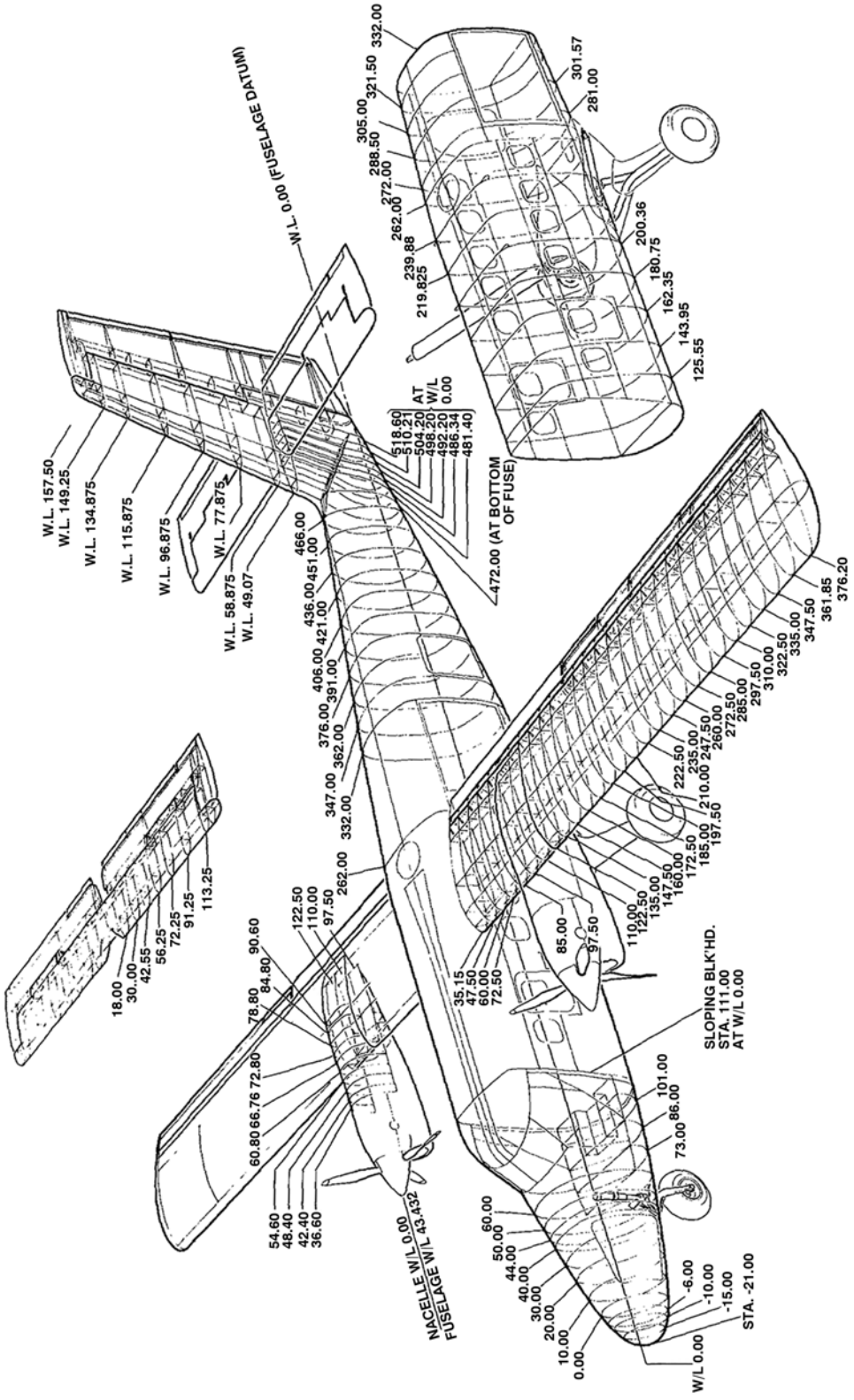
Main Wheels track	12 ft – 2 in.



Aircraft Dimensions
Figure 1

2. Aircraft Stations Diagram

In Figure 2, the stations diagram shows the location of the structural members of the aircraft in inches from three reference sources, each designated as station 0. Station 0 of the fuselage is 60.00 inches forward of the flight compartment front bulkhead, station 0 of the wing, horizontal stabilizer and elevator is the centerline of the aircraft; and station 0 of the vertical stabilizer and rudder is the aircraft datum line.



Aircraft Stations Diagram
Figure 2

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3. Symmetry and Alignment Checks

Symmetry and alignment checks are required if the aircraft has suffered a heavy landing, sustained damage or any of the major airframe components (wings, horizontal tail, vertical tail, fuselage) have been replaced. Complete the following symmetry and alignment checks. See Figure 3 for definition of points A, B, C, D, E, F and G. A difference of ± 0.5 inches from the production symmetry and alignment values should be taken as a warning of possible damage. Inspection of the relevant parts should be carried out to determine the extent of the damage.

Note Measurement techniques, measurement equipment and temperature may affect measurement accuracy; consult with Viking Air Limited Engineering if the above tolerances are exceeded.

A. Verify alignment by measuring the following distances per Figure 3:

1) Point A to B (37' – 2.5" ± 1.0") (_____)

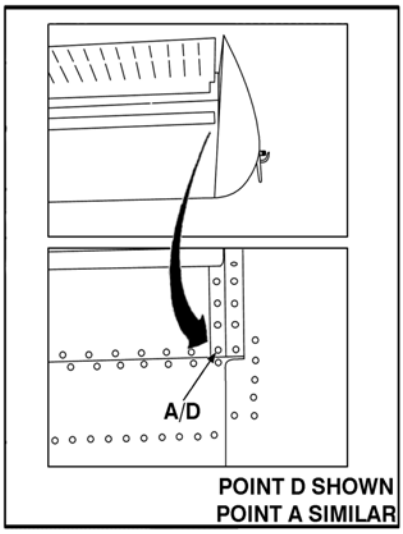
2) Point A to C (32' – 3.0" ± 1.0") (_____)

3) Point A to G (32' – 6.0" ± 1.0") (_____)

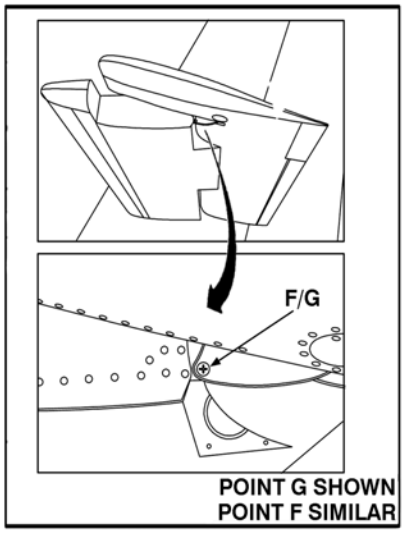
4) Point D to B (37' – 2.5" ± 1.0") (_____)

5) Point D to E (32' – 3.0" ± 1.0") (_____)

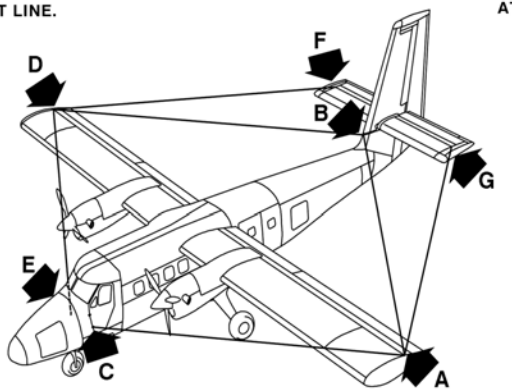
6) Point D to F (32' – 6.0" ± 1.0") (_____)

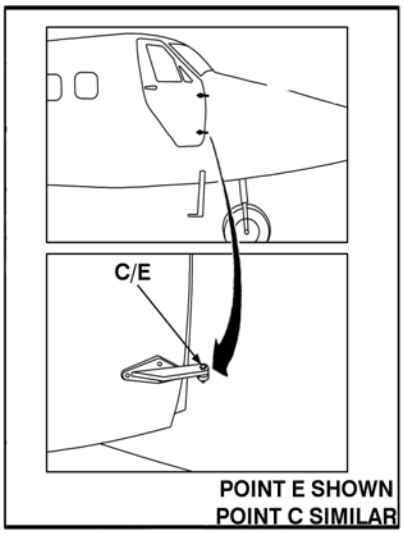


NOTE: VIEW LOOKING FORWARD, BOTTOM OF WING TIP, REAR SPAR RIVET LINES. POINT A AND D ARE THE FARTHEST OUTBOARD RIVETS ON THE REAR SPAR AFT RIVET LINE.

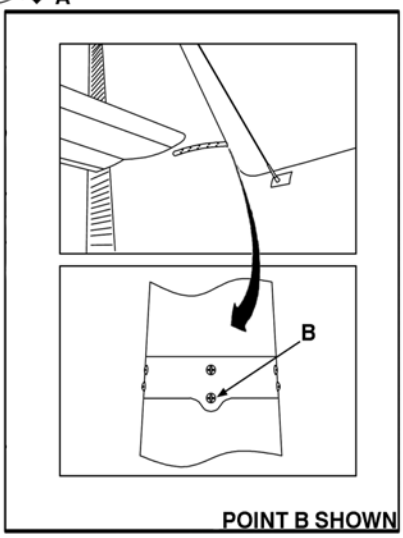


NOTE: VIEW LOOKING INBOARD ON HORIZONTAL TAIL (LEFT SIDE). POINT F AND G ARE THE OUTBOARD ELEVATOR HINGE LINE ATTACHMENT BOLTS.





NOTE: VIEW LOOKING AT RIGHT SIDE FORWARD COCKPIT DOOR. POINT C AND E (LEFT SIDE) ARE LOWER DOOR HINGE ATTACHMENT BOLTS.



NOTE: VIEW LOOKING AFT ON THE VERITCAL TAIL. POINT B IS THE BOTTOM SCREW ON THE VERTICAL TAIL/FUSELAGE JOINT FAIRING.

Symmetry and Alignment Checks
Figure 3